GROUP WORK PROJECT # 2 MScFE 622: Stochastic Modeling

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Statement of integrity

By typing the names of all group members below, the submission confirms that the assignment is original work produced by the group, excluding any member explicitly marked as non-contributing above.

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VIX Regime Modeling and State-Dependent ETF Allocation

A Markov Chain and Hidden Markov Model Analysis

VIX Regime Modeling and State-Dependent ETF Allocation

MScFE 622 Stochastic Modeling - Group Work Project #2

Abstract

This paper reproduces the analytical content of the executed project notebook in paper form. The common daily sample contains 5,465 complete observations from 19 November 2004 through 17 August 2026 for TLT, GLD, SPY, and VIX. ETF prices are converted to daily log returns and VIX to first differences. Two- and three-state discretized Markov chains and Gaussian hidden Markov models are estimated. State count is selected by BIC only within each likelihood family. The HMM K=3 candidate wins within the HMM family but fails the fixed minimum decoded-state occupancy diagnostic because its highest-mean state contains only 4.74% of observations; the deterministic fallback therefore selects Markov K=2. Historical state-conditional means map State 0 to SPY and State 1 to TLT. With the assignment-required one-observed-row execution lag, the rotation earns 2.88% annualized with a Sharpe ratio of 0.257, below both the monthly-reset equal-weight benchmark (8.95%, Sharpe 0.932) and SPY buy-and-hold (11.10%, Sharpe 0.652). Because regimes, model selection, and allocation means are estimated on the full sample, the backtest is descriptive and in-sample rather than causal evidence of live-trading profitability.

Introduction

The VIX is designed as an approximately 30-day forward-looking measure of expected S&P; 500 volatility inferred from SPX option quotations. The project asks whether changes in that volatility index can define useful market states for allocation among three economically distinct ETFs: TLT, GLD, and SPY. State labels remain numeric throughout because the state variable is the daily change in VIX rather than the VIX level itself; a positive-change state must not automatically be called a high-VIX or crisis regime.

Step 1 - Data Preparation and Exploration

The pipeline downloads Yahoo Finance Adjusted Close series for TLT, GLD, SPY, and ^VIX and retains only dates present for all four instruments. Missing common dates are removed before lags are formed; there is no filling or interpolation. The first lagged row is removed because it has no preceding common-sample observation.

$$r(i,t) = \ln(P(i,t) / P(i,t-1))$$

$$\text{Delta } VIX(t) = VIX(t) - VIX(t-1)$$

Delta VIX is a level change in VIX points, not a percentage return. It measures how the option-implied volatility index changes between consecutive rows of the common sample.

Canonical sample output

Statistic	Value
Start date	2004-11-19
End date	2026-08-17
Observations	5,465
Missing values	0

Table 1. Canonical common-sample metadata.

Date	TLT	GLD	SPY	VIX	TLT log r	GLD log r	SPY log r	Delta VIX
2004-11-19	43.302605	44.779999	78.858780	13.50	-0.008012	0.008973	-0.011178	0.520000
2004-11-22	43.528297	44.950001	79.234833	12.97	0.005198	0.003789	0.004757	-0.530000
2004-11-23	43.582287	44.750000	79.355743	12.67	0.001240	-0.004459	0.001525	-0.300000
2004-11-24	43.582287	45.049999	79.543793	12.72	0.000000	0.006682	0.002367	0.050000
2004-11-26	43.297714	45.290001	79.483330	12.78	-0.006551	0.005313	-0.000760	0.059999

Table 2. First five transformed observations displayed in the notebook.

The notebook visualizes the ETF log-return series and the daily VIX-change series over the full sample. The central empirical feature is that changes are centered near zero but exhibit occasional large moves. Those daily VIX changes, not the absolute VIX level, are the sole regime input in Step 2.

Step 2 - Modeling VIX Regimes: Discrete Markov Chains

For each requested state count K in {2,3}, the empirical distribution of daily VIX changes is divided by linear quantiles. An observation exactly equal to an interior cut is assigned to the higher-numbered state. This creates an observable finite-state Markov representation whose transition probabilities are estimated from consecutive state counts.

Markov candidate: K=2

The K=2 threshold is -0.0999994 VIX points. State 0 contains changes below the cut; State 1 contains changes at or above it. The stationary distribution is approximately balanced: 0.4987 for State 0 and 0.5013 for State 1.

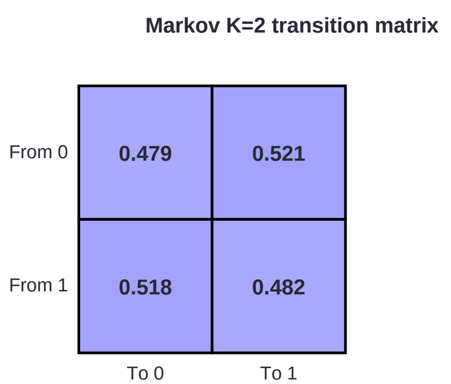

Figure 1. Selected Markov K=2 transition probabilities.

From state	To State 0	To State 1
0	0.4793	0.5207
1	0.5181	0.4819

Table 3. Markov K=2 transition matrix.

Markov candidate: K=3

The K=3 quantile cuts are -0.4700003 and +0.3000002 VIX points. The states therefore partition low, middle, and high daily VIX changes, while retaining neutral numeric labels. Stationary probabilities are 0.3322, 0.3336, and 0.3342.

From state	To State 0	To State 1	To State 2
0	0.3493	0.3008	0.3499
1	0.2331	0.4317	0.3352
2	0.4140	0.2683	0.3176

Table 4. Markov K=3 transition matrix.

These Markov constructions are descriptive and full-sample. Quantile thresholds are estimated on the full Step 1 sample, the first-order Markov property is imposed as an approximation, discretization discards within-state variation, and transition probabilities are assumed time-homogeneous.

Step 2 - Modeling VIX Regimes: Gaussian Hidden Markov Models

The HMM specification treats the regime as an unobserved finite-state process and models observed daily VIX change conditional on that latent state with a Gaussian distribution. Both K=2 and K=3 are fitted using five deterministic restarts with seeds 42-46, diagonal covariance, at most 500 EM iterations, convergence tolerance 10^-6, and minimum covariance 10^-6. Only converged finite-likelihood restarts are eligible. The highest log-likelihood fit is retained, with the lower seed breaking an effectively exact tie. State-dependent outputs are relabeled by increasing fitted mean Delta VIX.

The full-sample most-likely path is decoded with the Viterbi algorithm and smoothed posterior state probabilities are retained for every observation. These full-sample objects use observations after date t when classifying date t and therefore are not causal trading-time signals.

HMM candidate diagnostics

K	State	Mean Delta VIX	Variance	Viterbi obs.	Occupancy	Mean posterior
2	0	-0.0744	0.6543	4159	76.10%	0.7394
2	1	0.2126	11.9575	1306	23.90%	0.2606
3	0	-0.0621	0.3841	2900	53.06%	0.5218
3	1	-0.0114	3.1188	2306	42.20%	0.4211
3	2	0.6595	36.3474	259	4.74%	0.0571

Table 5. HMM state diagnostics. The HMM K=3 State 2 decoded occupancy of 4.74% is below the project 5% minimum.

For K=2, the retained restart is seed 45 with log-likelihood -9306.004199. For K=3, the retained restart is seed 46 with log-likelihood -8959.534974. The K=3 state means are approximately -0.0621, -0.0114, and +0.6595 VIX points. The sparse third state captures large upward VIX changes but occurs only 259 times.

Step 3 - Model Selection and Selected-State Provenance

Step 3 compares the two- and three-state candidates within each model family using log-likelihood, Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC). The Markov likelihood is the conditional likelihood of the observed transition sequence, whereas the Gaussian HMM likelihood is an observation-sequence likelihood after integrating over latent states. These are different likelihood objects. Consequently, raw AIC and BIC values are not ranked across Markov and HMM families.

$$AIC = -2 \ln(L) + 2k$$

$$BIC = -2 \ln(L) + k \ln(n)$$

Family	K	Log likelihood	Parameters	Obs.	AIC	BIC
Markov	2	-3783.223	2	5464	7570.4	7583.7
Markov	3	-5913.884	6	5464	11839.8	11879.4
HMM	2	-9306.004	7	5465	18626.0	18672.3
HMM	3	-8959.535	14	5465	17947.1	18039.6

Table 6. Four-candidate information-criterion table. criterion_scope = within_family_only.

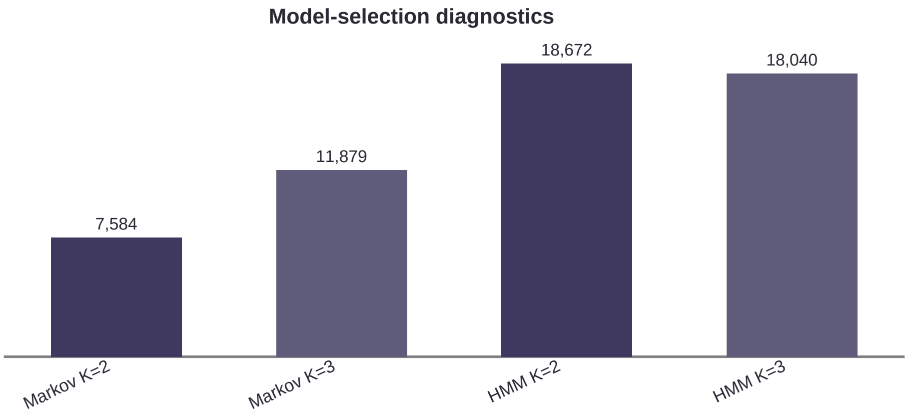

Figure 2. BIC values are interpreted only within each likelihood family.

Selected model: MARKOV with K=2. Within-family BIC selects HMM K=3, but at least one decoded HMM state has occupancy below 5%; the fixed validity rule therefore falls back to Markov K=2. The selected Date/state series is persisted rather than re-estimated later, providing an auditable provenance chain into the conditional ETF analysis and allocation steps.

The numeric labels are retained: State 0 corresponds to the lower daily VIX-change partition and State 1 to the higher partition. They do not represent absolute low/high VIX levels.

Step 3 - State-Conditional ETF Analysis

The project loads the canonical selected-state series, requires exact date alignment with the frozen Step 1 dataset, and computes each ETF daily mean log return, sample standard deviation, and observation count within each selected state. The regime models are not refitted and the model-selection decision is not revisited.

State	Asset	Mean log return	Std. log return	Observations
0	TLT	-0.00125743	0.00885398	2725
0	GLD	0.00086016	0.01087795	2725
0	SPY	0.00670184	0.00933695	2725
1	TLT	0.00147775	0.00924862	2740
1	GLD	-0.00004804	0.01205600	2740
1	SPY	-0.00583631	0.01084325	2740

Table 7. State-conditional ETF statistics from the selected Markov K=2 state path.

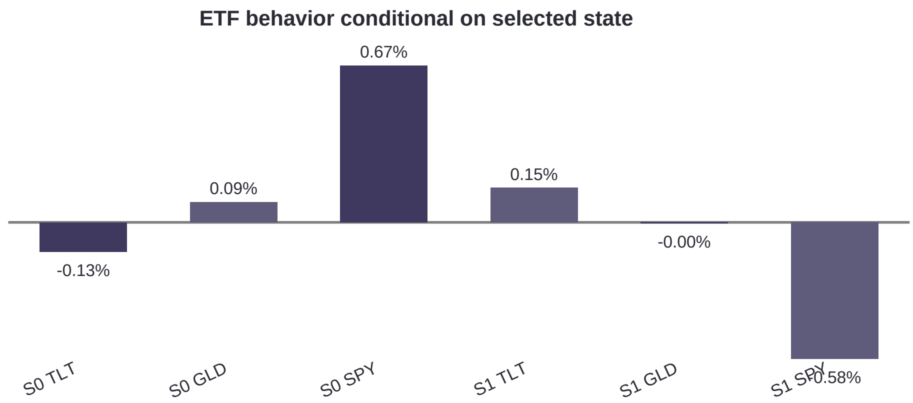

Figure 3. State-conditional mean daily log returns.

State 0 is strongly favorable to SPY in-sample: mean daily log return 0.00670184, compared with 0.00086016 for GLD and -0.00125743 for TLT. In State 1, TLT leads at 0.00147775, while SPY averages -0.00583631. These full-sample conditional means are diagnostic evidence for Step 4 and can introduce optimistic selection bias if the same dates are subsequently used for evaluation.

Step 4 - State-Based Rotation Rule

For each selected state, the strategy assigns 100% weight to the ETF with the largest historical conditional mean daily log return. Exact ties are broken by the fixed priority TLT -> GLD -> SPY. The optional 60/40 variant is not used.

State	Selected asset	Selection mean log return	TLT weight	GLD weight	SPY weight
0	SPY	0.00670184	0.0	0.0	1.0
1	TLT	0.00147775	1.0	0.0	0.0

Table 8. Deterministic allocation mapping.

The mapping is also in-sample: regime thresholds/model parameters, selected states, and conditional ETF means are all estimated from the full analysis sample. The later execution lag addresses same-row timing only and does not make the mapping causal or out-of-sample.

Step 5 - Backtest Construction and Required Benchmarks

ETF log returns are converted to simple returns before portfolio arithmetic. The Step 4 allocation is applied with an exact one-observed-trading-row lag: the state on row t-1 determines the weights applied to ETF returns on row t. The first Step 1 row is excluded because no preceding regime decision exists.

$$R(i,t) = \exp(r(i,t)) - 1$$

The equal-weight benchmark starts at one-third TLT, one-third GLD, and one-third SPY, resets to those weights immediately before the first comparison return and before the first observed return in each new calendar month, and allows weights to drift inside the month. The second benchmark is SPY buy-and-hold. All three portfolios use identical comparison dates. Reported returns are gross of transaction costs and taxes; the implemented cost assumption is zero.

Performance metrics and cumulative comparison

Performance uses simple daily portfolio returns, 252 trading days per year, a zero risk-free rate, sample volatility with denominator n-1, and initial wealth W0=1.

$$W(t) = W(t-1)\,[1 + R_p(t)]$$

$$\text{Annualized return} = (W(T)/W(0))^{(252/n)} - 1$$

$$\text{Annualized volatility} = sd(R_p)\,\sqrt{252}$$

$$\text{Sharpe} = mean(R_p) / sd(R_p)\,\sqrt{252},\ \text{with } r_f = 0$$

$$\text{Maximum drawdown} = \min_t \left[W(t) / \max_{(s<=t)} W(s) - 1 \right]$$

Portfolio	Cumulative	Annualized	Ann. vol.	Sharpe	Max DD	Obs.
Regime rotation	0.849021	0.028754	0.160369	0.257204	-0.537600	5464
Equal-weight monthly	5.420849	0.089548	0.097112	0.931856	-0.230437	5464
SPY buy-and-hold	8.798148	0.110994	0.188845	0.651956	-0.551894	5464

Table 9. Exact notebook performance summary on 5,464 identical lagged comparison dates.

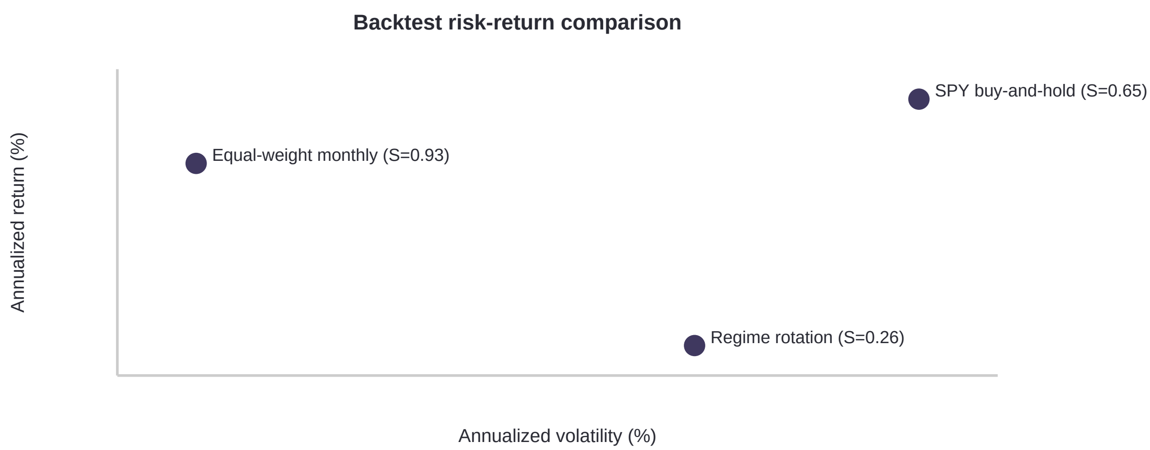

Figure 4. Annualized risk-return position of the strategy and required benchmarks.

The regime rotation returns 0.849021 cumulatively, 0.028754 annualized, with 0.160369 annualized volatility, Sharpe 0.257204, and maximum drawdown -0.537600. The monthly-reset equal-weight portfolio reaches 5.420849 cumulative return, 0.089548 annualized, volatility 0.097112, Sharpe 0.931856, and maximum drawdown -0.230437. SPY buy-and-hold reaches 8.798148 cumulative and 0.110994 annualized, with volatility 0.188845, Sharpe 0.651956, and maximum drawdown -0.551894.

The regime rotation therefore does not outperform either required benchmark. Equal-weight monthly reset has the best risk-adjusted performance and shallowest drawdown, while SPY has the highest total and annualized return.

Interpretation, Provenance, and Limitations

The notebook and this paper should be read as a full-sample descriptive study. For the Markov models, quantile thresholds are estimated from the full Step 1 sample, the Markov property is an imposed approximation, discretization discards within-state variation, and transition probabilities are assumed time-homogeneous. The selected K=2 chain is not particularly persistent: self-transition probabilities are only about 0.48.

For the HMMs, deterministic restarts reduce but do not eliminate local-optimum risk in EM. More importantly, the Viterbi state path and smoothed posterior probabilities use the full observation sequence. If they were used as trading signals, future information would enter historical classifications. The project avoids selecting the sparse HMM K=3 path by means of a fixed occupancy diagnostic, but that diagnostic does not make any HMM full-sample classification causal.

The largest limitation for the portfolio test is parameter-estimation look-ahead. The state definitions/model parameters, model-selection decision, selected state path, and state-conditional ETF means are all produced from the same full historical sample that is later backtested. The one-row execution lag prevents using the same row state to trade the same row return, but it cannot undo full-sample estimation or allocation-selection bias. The results are therefore descriptive rather than evidence of live-trading profitability.

A stronger research design would estimate regimes and state-conditional allocations using rolling or expanding windows; use filtered HMM probabilities available at each decision date; reserve an untouched test period; include turnover, bid-ask spreads, commissions, and taxes where relevant; and compare the concentrated winner-take-all rule with diversified or risk-controlled state-dependent weights.

The concentrated rule also ignores diversification benefits emphasized by Markowitz (1952). Repeated model/strategy selection on the same sample can inflate apparent performance through data snooping and backtest overfitting, motivating the cautions discussed by White (2000) and Bailey and Lopez de Prado (2014).

Conclusion

The project constructs a reproducible VIX-change regime workflow from data preparation through regime estimation, model selection, allocation, and benchmarked backtesting. The deterministic selection rule chooses Markov K=2 after the lower-BIC HMM K=3 candidate fails the 5% decoded-state occupancy requirement. The selected state mapping allocates to SPY in State 0 and TLT in State 1. Despite the intuitive directional mapping, the full-sample rotation materially underperforms the equal-weight monthly-reset portfolio and SPY buy-and-hold.

The main contribution is methodological rather than performance-based: transparent state definitions, within-family information-criterion comparison, explicit method-validity diagnostics, persistence of the selected state source, and an execution lag make the workflow auditable. The weak benchmark performance and the full-sample design also show why regime classification alone does not establish a profitable trading strategy. The necessary next step is genuinely out-of-sample, decision-time-only validation.

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Yahoo Finance. Daily adjusted close market observations for TLT, GLD, SPY, and ^VIX, downloaded by the project loader.

Appendix A - Selected Candidate Diagnostics

Selected Markov K=2 threshold: -0.0999994278 VIX points. Stationary probabilities: State 0 = 0.4987188873; State 1 = 0.5012811127.

Within-family winners: Markov K=2 by BIC 7583.658 vs 11879.403; HMM K=3 by BIC 18039.556 vs 18672.251. HMM K=3 fails method validity because State 2 Viterbi occupancy is 259/5465 = 4.739%, below 5%.

Final state source: reports/tables/step2_markov_2_states.csv, persisted downstream as reports/tables/step3_selected_states.csv. Allocation mapping: State 0 -> SPY; State 1 -> TLT. Backtest observations after one-row lag: 5,464.